Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/18/2016
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mechanistic Analysis of Particle Radiation-Induced Carcinogenesis Using Validated Mouse Glioma Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	burma@uthscsa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	210-450-8981
Organization Name:	University of Texas Health San Antonio		
PI Address 1:	Department of Neurosurgery, Department of Biochemistry and Structural Biology		
PI Address 2:	7703 Floyd Curl Drive, MC7843, Room 102F		
PI Web Page:			
City:	San Antonio	**State:**	TX
Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Flight	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	01/20/2016	**End Date:**	01/19/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX16AD78G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Galactic Cosmic Rays (GCR) and Solar Particle Events (SPEs) are a major source of carcinogenic risk for astronauts on prolonged space missions. Large uncertainties exist in the exact estimation of cancer risks from charged particles due to the paucity of epidemiological data in this area. Furthermore, a mechanistic understanding of genetic changes underlying transformation by particle radiation is necessary for the development of appropriate countermeasures. Thus, experimental animal models that closely mimic the process of carcinogenesis in humans are essential for oncogenic risk assessment. In the course of our currently funded 3 yr. NASA project (2013–2016), we characterized two sensitive and complementary mouse models that can be used to elucidate molecular mechanisms underlying the process of particle radiation-induced glioblastoma (GBM) development. GBMs are lethal brain tumors with very dismal prognosis for which radiation is the only known risk factor. GBMs represent the third leading cause of cancer-related death among adults aged 30 – 50 years (the average age of astronauts), the mean survival after diagnosis being only about 14 months. GBM was the first cancer to be analyzed by The Cancer Genome Atlas Network (TCGA), and the key genetic alterations occurring in GBM are now well defined. Based on this information, we utilized transgenic mice with brain-restricted deletions of GBM-relevant tumor suppressors in logical combinations to analyze the process of charged particle-induced carcinogenesis. We identified and characterized two complementary mouse GBM models that would be ideal for studying particle radiation-induced carcinogenesis - Nestin-Cre; Ink4ab-/-Arf f/f and Nestin-Cre; p53f/+Ptenf/+. These models exhibit a low frequency of spontaneous GBMs but readily develop brain tumors after exposure to HZE (high energy) particles, which resemble human high grade gliomas in their genetic and molecular signatures. Using these mice, we have clearly demonstrated that high-linear energy transfer (LET) charged particles have a greater transforming potential compared to low-LET radiation, and have characterized the molecular events underlying Fe ion-induced gliomagenesis in these models. This work sets the stage for quantitative studies on radiation quality effects on carcinogenesis for an array of charged particles representing the GCR (as proposed in Aim 1). These charged particles induce complex DNA double-strand breaks (DSBs) the accurate repair of which is critical for preventing transformation. Our models will also allow us to determine which DSB repair pathway represents a major barrier to charged particle-induced gliomagenesis (as proposed in Aim 2). This information would be fundamental to the development of a mechanistic understanding of charged particle-induced carcinogenesis. We will cross the transgenic mice characterized by us with mice harboring brain-specific ablation of non-homologous end joining (NHEJ) or homologous recombination (HR) repair pathways -- Nestin-Cre; Ligase4f/f and Nestin-Cre; Brca2f/f – that were established in the laboratory of Prof. Peter McKinnon (collaborator on project). Finally, genomic and bioinformatics analyses will be used to define genetic signatures unique to particle-radiation induced glioblastomas and to delineate key pathways that are de-regulated during HZE-induced gliomagenesis (as proposed in Aim 3). We are hopeful that an in-depth mechanistic analysis of charged particle-induced gliomagenesis in these validated mouse models will help us to make significant progress towards cancer risk assessments and development of protective strategies for charged particle exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth. Our examination of the relationship between LET and gliomagenesis could have practical implications for radiotherapy modality choice.
Task Progress:	The most comprehensive data set correlating linear energy transfer (LET) with carcinogenic potential comes from the mouse Harderian gland studies carried out in the 1980s. These studies indicate an initial linear relative biological effectiveness (RBE)-LET relationship that seems to plateau at higher LETs or doses. These and other in vitro studies indicate a maximal LET for cellular transformation at or around 100 keV/µm. As humans do not have Harderian glands, it is important to also use mouse models of human-relevant cancers to quantitatively assess radiation quality effects on carcinogenesis. The two complementary mouse models that we have characterized in the course of our previous 3-year NASA project (2013-2016) -- Nestin-Cre; Ink4ab-/-Arf f/f and Nestin-Cre; p53f/+Ptenf/+ -- are uniquely suited to further elucidate the role of radiation quality effects on charged particle-induced carcinogenesis. These models exhibit a low frequency of spontaneous glioblastomas (GBMs) but are susceptible to radiation-induced brain tumors which resemble human high grade gliomas histopathologically and molecularly. Indeed, exploratory studies with four different ion species (H, C, Si, and Fe) clearly show that charged particles possess a greater gliomagenic potential relative to low LET X-rays. As a continuation of this work, in the first year of our grant, we are evaluating the relative contributions of LET and atomic number (Z) to carcinogenesis and obtaining specific RBE values for representative charged particles using our two mouse models. Previously we found that irradiation with Si (Z=14; LET=79 keV/µm) resulted in the highest incidence of radiogenic gliomas while irradiation with heavier Fe ions (Z=26; LET=174 keV/µm) resulted in lower tumor prevalence. These results indicate that Si (or LETs of ~ 79 keV/µm) is the most detrimental with respect to carcinogenesis out of all the ions we tested. However, there is a substantial gap in the Z and LET values between Si and the other species (C and Fe) that we tested. Thus, additional species with Z and LET values that fall in between C and Si and between Si and Fe were selected to: a) confirm the initial linear response between LET and tumor frequency, and b) to establish that Si with LET of ~79 keV/µm is one of the most detrimental ions in terms of cancer risk. To fill-in this gap and substantiate our existing results we chose two additional ions: Ne (Z=10; LET=39 keV/µm) and Ti (Z=22; LET=120 keV/µm). During the NASA Space Radiation Laboratory (NSRL) 16B campaign (first campaign funded by NNX16AD78G) Nestin-Cre; Ink4ab-/-Arf f/f mice were irradiated with 0.5 Gy of Ne ions or Ti ions. Following IR, the mice were shipped back to UTSW (University of Texas Southwestern Medical Center) where we monitor them daily for neurologic symptoms of gliomagenesis until the end of the study (9 month post IR). Complete information about Ne-induced gliomagenesis will not be available until February 2017, but we have already observed multiple GBM cases, and Ne appears to be more gliomagenic than C. Additional 20 animals need to be irradiated with Ti during NSRL 17A to complete the cohort. Out of the 20 mice irradiated with Ti during NSRL 16B, multiple mice have already developed GBMs. Final GBM development frequencies for Ne and Ti will become available by February 2017. In addition, in order to understand the effects of LET versus atomic number (Z) on tumor frequency, we irradiated multiple cohorts of Nestin-Cre; Ink4ab-/-Arf f/f mice with Si ions of different energies and LETs during the NSRL 16C campaign. Gliomagenesis data from these cohorts will be available by July 2017. Complementary experiments, where we will use three different ion species (Ne, Si, and Cl) but with the same LET of 79 keV/µm (most tumorigenic as per our current results), are planned for 2017.

Bibliography Type:	Description: (Last Updated: 12/09/2025)
Articles in Peer-reviewed Journals	Gil Del Alcazar CR, Todorova PK, Habib AA, Mukherjee B, Burma S. "Augmented HR repair mediates acquired temozolomide resistance in glioblastoma." Mol Cancer Res. 2016 Oct;14(10):928-40. http://dx.doi.org/10.1158/1541-7786.MCR-16-0125 ; PubMed PMID: 27358111; PubMed Central PMCID: PMC5065752 , Oct-2016
Articles in Peer-reviewed Journals	Morris C, Tomimatsu N, Burma S, Jalinot P. "INT6/EIF3E controls the RNF8-dependent ubiquitylation pathway and facilitates DNA double-strand break repair in human cells." Cancer Res. 2016 Oct 15;76(20):6054-65. http://dx.doi.org/10.1158/0008-5472.CAN-16-0723 ; PubMed PMID: 27550454; PubMed Central PMCID: PMC5065779 , Oct-2016
Journal/Magazine covers	Gil Del Alcazar CR, Todorova PK, Habib AA, Mukherjee B, Burma S. "Cover in journal Cancer Research for article, 'Augmented HR repair mediates acquired temozolomide resistance in glioblastoma.' " Mol Cancer Res. 2016 Oct;14(10):928-40. http://dx.doi.org/10.1158/1541-7786.MCR-16-0125 ; PubMed PMID: 27358111; PubMed Central PMCID: PMC5065752 ; Cover: http://mcr.aacrjournals.org/content/14/10.cover-expansion , Oct-2016